

ALG-03: Polynomials, Roots, Vieta & Remainders

Student Assimilation Pack | Representation before calculation

RECONNECT → DISCOVER → MAKE SENSE → TRY → DIAGNOSE → FADE → ADOPT →
TRANSFER

Polynomials, Roots, Vieta & Remainders

Integrated Assimilation Book

> The requested information chooses the representation.

1. RECONNECT - one polynomial, several useful forms

Consider

$$P(x)=x^2-5x+6.$$

Expanded form exposes coefficients:

$$x^2-5x+6.$$

Factored form exposes zeros:

$$(x-2)(x-3).$$

Root list exposes the same zero information directly:

$$2,3.$$

No representation is universally best.

First attempt

For each request, name the cheapest representation before solving.

1. Find the roots of x^2-5x+6 .
2. Find the sum and product of the roots of $2x^2-5x-3$.
3. Decide how many real roots $x^2-4x+5=0$ has.
4. Find the remainder of x^{100} when reduced under $x^2=1$.

2. DISCOVER - roots and factors are the same zero structure

If $P(r)=0$, then $x-r$ is a factor of $P(x)$.

For a quadratic with roots **alpha,beta** and leading coefficient **a**,

$$P(x)=a(x-\text{alpha})(x-\text{beta}).$$

Expanding,

$$P(x)=a[x^2-(\text{alpha}+\text{beta})x+\text{alpha beta}].$$

Compare this with

$$ax^2+bx+c.$$

Coefficient matching gives

$$-a(\text{alpha}+\text{beta})=b, \text{ so}$$

$$\text{alpha}+\text{beta}=-b/a,$$

and

$$a \text{ alpha beta}=c, \text{ so}$$

$$\text{alpha beta}=c/a.$$

This is Vieta derived from factor expansion. It is not an independent formula list.

Use the information the target asks for

For $x^2-7x+10=0$, if the target is $\text{alpha}^2+\text{beta}^2$, do not solve roots first:

$$\text{alpha}+\text{beta}=7, \text{ alpha beta}=10,$$

so

$$\text{alpha}^2+\text{beta}^2=49-20=29.$$

Contrast

- “find **alpha,beta**” -> individual roots may be needed;
- “find **alpha^2+beta^2**” -> symmetric invariants are enough.

3. MAKE SENSE - discriminant describes quadratic root behavior

For

$$ax^2+bx+c=0, a \neq 0,$$

the quadratic formula is

$$x=(-b \pm \sqrt{b^2-4ac})/(2a).$$

The quantity

$$\Delta=b^2-4ac$$

controls real-root behavior:

- **Delta>0**: two distinct real roots;
- **Delta=0**: one repeated real root;
- **Delta<0**: no real roots.

Root geometry

The parabola $y=ax^2+bx+c$ meets the x-axis twice, touches once, or does not meet it according to the same three cases.

Mandatory boundary with ALG-02

Question: “How many real roots does $x^2-4x+5=0$ have?” Use discriminant: **Delta=16-20=-4**, so none.

Question: “What is the minimum value of x^2-4x+5 ?” That is an optimization request. Complete the square in ALG-02: $(x-2)^2+1$, minimum **1**.

Same quadratic, different requested information, different canonical method.

4. DISCOVER - transformed roots and shifted input move in opposite directions

Suppose **P(x)** has roots **alpha,beta**.

To build a polynomial with roots **alpha+c, beta+c**, replace **x** by **x-c**:

$$Q(x)=P(x-c).$$

Why? If $x=\alpha+c$, then $x-c=\alpha$, so $P(\alpha)=0$.

Example

P(x)=x^2-5x+6 has roots **2,3**.

Roots shifted up by 4 are **6,7**.

Use

$$Q(x)=P(x-4)=(x-6)(x-7)=x^2-13x+42.$$

But **P(x+4)** has roots **-2,-1**.

> Root shift **+c** requires input shift **-c**.

5. MAKE SENSE - remainder theorem and factor theorem

When a polynomial **P(x)** is divided by **x-a**, the remainder is **P(a)**.

Reason: division gives

$$P(x)=(x-a)Q(x)+r,$$

where **r** is constant. Set $x=a$:

$$P(a)=r.$$

Therefore:

- remainder on division by $x-a$ is $P(a)$;
- $x-a$ is a factor exactly when $P(a)=0$.

Example

For $P(x)=x^3+2x+5$, remainder on division by $x-2$ is
 $P(2)=8+4+5=17$.

6. DISCOVER - polynomial reduction is the canonical extension of ALG-01 relation rewriting

ALG-01 taught the local habit: if an admissible x satisfies $x^2=3x-1$, replace higher powers rather than solving the roots.

ALG-03 generalizes this to polynomial reduction.

Modulo the relation

$$x^2-3x+1=0,$$

we may write

$$x^2 \equiv 3x-1.$$

Every higher power reduces to degree less than 2.

For example,

$$x^3 \equiv x(3x-1) = 3x^2-x \equiv 8x-3,$$

and ultimately

$$x^5 \equiv 55x-21.$$

The language is now about the remainder class modulo a polynomial, not only values at individual roots.

Contrast

To find x^{2026} modulo x^2+x+1 , do not calculate 2026 multiplications.

Because

$$x^3-1=(x-1)(x^2+x+1),$$

we have

$$x^3 \equiv 1 \pmod{x^2+x+1}.$$

Since $2026 \equiv 1 \pmod{3}$,

$$x^{2026} \equiv x.$$

7. MAKE SENSE - common-root elimination lowers degree

Suppose a number is a common root of

$$x^2-5x+6=0$$

and

$$x^2-4x+3=0.$$

Subtract the equations:

$$-x+3=0.$$

So any common root must be $x=3$.

Check it in both originals: it works.

This is cheaper than solving two quadratics separately and intersecting the answer sets.

8. TRY - H3 -> H2 -> H1 -> H0

Attempt before support.

H3 - execution relation supplied

Roots **alpha,beta** of $x^2-8x+12=0$. Find $\alpha^2+\beta^2$.

Use Vieta **alpha+beta=8**, **alpha beta=12**, then $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=40$.

H2 - representation supplied

Find the polynomial whose roots are **5** more than the roots of x^2-3x-4 .

Cue: use **P(x-5)** and simplify.

H1 - recognition clue

Find the remainder of x^{50} modulo x^2-1 .

Clue: do not expand powers; reduce the relation first.

H0 - no route supplied

Two monic quadratics share a root: $x^2-7x+10=0$, $x^2-5x+4=0$. Find the common root.

9. DIAGNOSE - four representation errors

Error A - solve roots when target is symmetric

Repair: Vieta first; solve roots only if target still distinguishes them.

Error B - use discriminant for a vertex-value request

Repair: root count -> ALG-03 discriminant; minimum/maximum -> ALG-02 square completion.

Error C - shift polynomial input in the same direction as desired root shift

Repair: test one root. Root **alpha+c** must make the input equal **alpha**, hence **P(x-c)**.

Error D - calculate high powers directly

Repair: identify the polynomial relation and reduce after each multiplication, or exploit periodicity.

10. ADOPT - one topic-wide First-Step logic

REQUEST

-> roots individually? -> factor/solve

-> symmetric root information? -> Vieta

-> root count/behavior? -> discriminant

-> transformed roots? -> transform input

-> remainder/factor? -> evaluate or divide

-> high power modulo relation? -> reduce

-> common root? -> eliminate

-> CHECK

11. Historical anchor traces

IOQM-2025-Q16

Combine the given polynomial functions so the root information is visible. The independently verified route reconstructs a quadratic with roots **-2,3**, determines its scale, and recovers the requested value **22**.

IOQM-2025-Q24

Factor the divisor as $(x^2+1)(x^2+x+1)$ and reduce x^{2025} modulo the polynomial instead of computing the high power. The independently verified remainder evaluated at the requested point gives **53**.

IOQM-2024-Q24

The coefficient restriction creates a finite structural polynomial problem; exhaustive independent checking of the admissible binary-coefficient configurations confirms answer **50**.

IOQM-2023-Q12

A cubic identity creates a factorized alternative; parity/admissibility eliminates one branch and the remaining equal-structure branch gives answer **18**.

These anchors are mechanism evidence, not weightage evidence.

12. TRANSFER

- symmetric root expressions -> algebraic invariants without explicit roots;
- parameter root-count questions -> discriminant feasibility;
- recurrence/high-power problems -> polynomial reduction / periodic remainders;
- common-root systems -> elimination/GCD thinking;
- transformed-root questions -> input transformation, a useful precursor to functional transformations.

Final belief

> Before solving a polynomial, I ask what information the target needs and choose the representation that exposes exactly that information.

ALG-03 - First-Step Reference

One topic-wide router

REQUEST

- > individual roots? FACTOR / SOLVE
- > symmetric root target? VIETA
- > root count/real behavior? DISCRIMINANT
- > shifted/transformed roots? TRANSFORM INPUT
- > remainder/factor? EVALUATE / DIVIDE
- > high power modulo relation? REDUCE
- > common root? ELIMINATE
- > CHECK

Recognition atlas

Target clue	First question
alpha+beta , alpha beta , symmetric powers	do coefficients already determine it?
number of real roots / repeated root	what is Delta ?
roots shifted by c	should polynomial be P(x-c) ?
divisor x-a	evaluate P(a) ?
huge exponent + polynomial relation	what low-degree remainder class repeats?
two equations share root	can subtraction/combination lower degree?

Four mandatory contrasts

- solve roots vs use symmetric invariants;
- discriminant/root-count vs vertex/minimum;
- transformed roots vs shifted input;
- calculate high power vs reduce modulo a polynomial.

ALG-03 - Recognition and First-Line Lab

Write only the first useful line or representation decision.

1. Roots **α, β** of $x^2 - 7x + 10$; target **$\alpha^2 + \beta^2$** .
2. Find roots of $x^2 - 7x + 10$.
3. Count real roots of $x^2 - 4x + 5 = 0$.
4. Find minimum value of $x^2 - 4x + 5$.
5. Polynomial **P** has root **3** ; what factor follows?
6. Find remainder of $x^3 + 2x + 5$ on division by $x - 2$.
7. Roots of **P** are **$2, 3$** ; desired roots **$6, 7$** . Which input transform?
8. If **$Q(x) = P(x + 4)$** , how do its roots move?
9. Reduce x^{100} modulo $x^2 - 1$.
10. Reduce x^{2026} modulo $x^2 + x + 1$.
11. Two quadratics share a root. What operation can lower degree before solving?
12. Roots **r, s** of $2x^2 - 5x - 3$; target **$1/r + 1/s$** . What invariants are needed?
13. Determine whether $x + 1$ is a factor of $x^3 + 2x^2 - x - 2$.
14. A parameter question asks for a repeated real root. What canonical quantity controls it?
15. You need polynomial with roots **$2\alpha, 2\beta$** where **$P(\alpha) = P(\beta) = 0$** . What substitution pattern should be tested?
16. Given a huge exponent but no polynomial relation/divisor, can polynomial reduction be invoked yet? State what information is missing.

ALG-03 - Practice and Transfer Bank

F0 - foundation

1. Factor x^2-5x+6 and list its roots.
2. For roots α, β of $x^2-7x+10$, find $\alpha+\beta$ and $\alpha\beta$ from factor expansion/Vieta.
3. Find $\alpha^2+\beta^2$ for the same roots without solving them individually.
4. Compute the discriminant of x^2-6x+9 and state root behavior.
5. Find the remainder of x^2+3x+4 on division by $x-1$.

F1 - direct

6. For roots of $2x^2-5x-3$, find their sum and product.
7. For roots of $x^2-8x+12$, find $\alpha^3+\beta^3$.
8. Determine root behavior of $x^2-4x+5=0$.
9. Determine root behavior of $x^2-5x+6=0$.
10. Test whether $x+1$ is a factor of x^3+2x^2-x-2 .

F2 - standard

11. $P(x)=x^2-5x+6$. Find the monic polynomial whose roots are each 4 greater than the roots of P .
12. What are the roots of $P(x+4)$ if P has roots 2,3?
13. Reduce x^{100} modulo x^2-1 .
14. Reduce x^{2026} modulo x^2+x+1 .
15. Under $x^2-3x+1=0$, reduce x^5 to $Ax+B$ and express it as a polynomial remainder statement.

F3 - disguised

16. Two equations $x^2-5x+6=0$ and $x^2-4x+3=0$ share a root. Find it by elimination.
17. Roots α, β of $3x^2+2x-5$. Find $1/\alpha+1/\beta$ without solving roots.
18. Roots p, q of $x^2-sx+t=0$. Express p^3+q^3 in s, t .
19. Find all k for which $x^2-kx+9=0$ has a repeated real root.
20. For $P(x)=x^3-4x+3$, find the remainder upon division by $x-1$, and infer whether $x-1$ is a factor.

F4 - preliminary-style

21. A monic quadratic has roots α, β , with $\alpha+\beta=9$ and $\alpha^2+\beta^2=41$. Find $\alpha\beta$ and the polynomial.
22. If x^2+x+1 divides a polynomial relation, reduce $x^{1000}+x^{999}+x^{998}$ modulo x^2+x+1 .
23. Polynomials x^2+ax+6 and x^2+bx+3 share root -1 . Determine a, b and their other roots.
24. Find the remainder of x^{20} upon division by x^2+x+1 .
25. A quadratic has integer coefficients, roots differing by 1, sum 9. Reconstruct the roots and monic polynomial using invariant information before expanding.

Transfer T2-T4

26. T2 representation: solve $\alpha^2+\beta^2$ once by explicit roots and once by Vieta; compare information cost.
27. T2 boundary: for x^2-4x+5 , answer both "number of real roots" and "minimum value," naming the two canonical topics/methods.
28. T3 transformed input: if roots of P are unknown but the target roots are $\alpha-3, \beta-3$, write the transformed polynomial directly.
29. T3 recurrence: explain why a recurrence characteristic relation can turn high powers into low-degree remainders without teaching the recurrence topic here.
30. T4 common structure: explain how common-root elimination is a precursor to polynomial gcd thinking.

ALG-03 - H0 Mixed Mastery Test

No labels or default hints.

1. Roots **alpha, beta** satisfy $2x^2 - 9x + 4 = 0$. Find $\alpha^2 + \beta^2$ without unnecessary root solving.
2. Determine the number of real roots of $3x^2 + 2x + 5 = 0$.
3. Determine the repeated root of $4x^2 - 12x + 9 = 0$.
4. $P(x) = x^2 - 6x + 8$. Find the polynomial whose roots are **3** more than the roots of **P**.
5. If $Q(x) = P(x-2)$, describe how every root of **P** moves to a root of **Q**.
6. Find the remainder of $x^4 + 2x + 7$ on division by $x-3$.
7. Determine whether $x-2$ is a factor of $x^3 - 3x^2 + 4$.
8. Reduce x^{99} modulo $x^2 + x + 1$.
9. Reduce x^6 modulo $x^2 - 2x - 1$.
10. Two quadratics $x^2 - 6x + 8 = 0$ and $x^2 - 5x + 6 = 0$ share a root. Find it without solving both completely.
11. Roots **r, s** of $x^2 - 10x + 21$. Find $r^3 + s^3$.
12. A question asks for the minimum value of $x^2 - 10x + 21$. Route it to the correct canonical topic and state the cheapest representation.
13. A question asks for the values of **k** giving two distinct real roots of $x^2 + kx + 4 = 0$. State the governing inequality.
14. WHY NOT: a student expands x^{100} before reducing modulo $x^2 - 1$. Diagnose the representation error.
15. WHY NOT: a student says roots shifted by **+5** come from $P(x+5)$. Refute by substituting one old root.
16. Transfer: two polynomials share a root and their difference is linear. Explain why solving the linear difference first is logically sufficient only after the candidate is checked in both originals.